VIT-AP UNIVERSITY, ANDHRA PRADESH

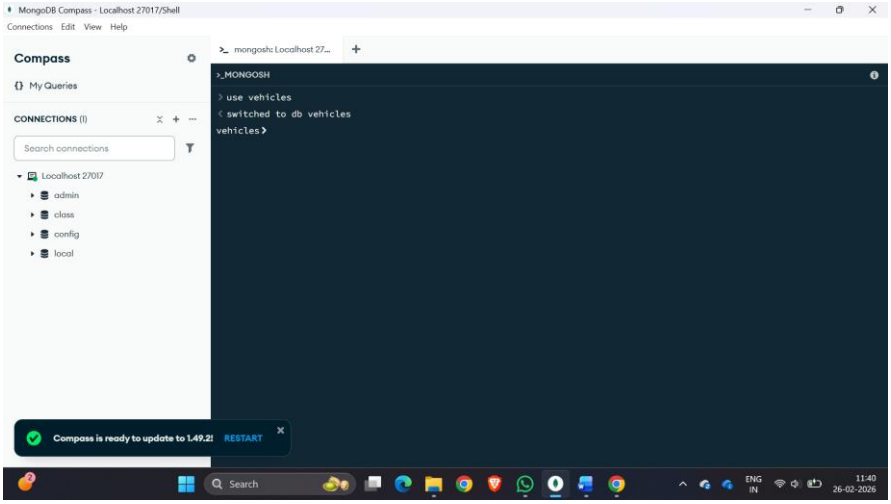
Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date:26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: P.Prasanth kumar **Reg. no.: 23BCE9742**

1. Use MongoDB to implement the following DB operations

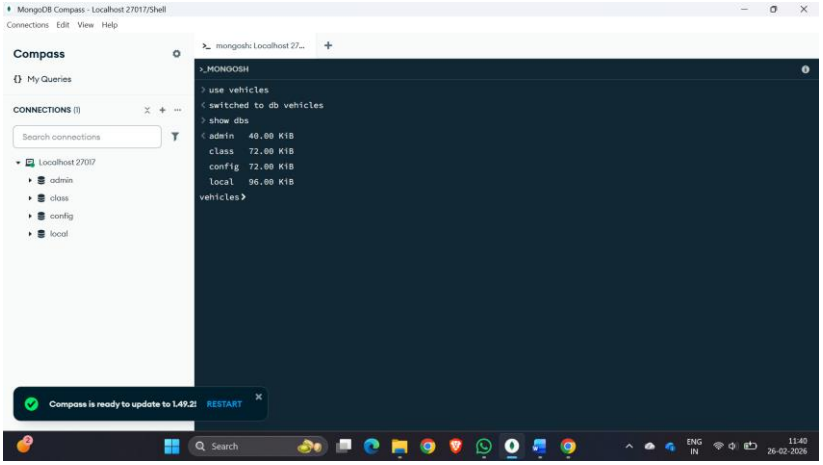
   1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

      output:-



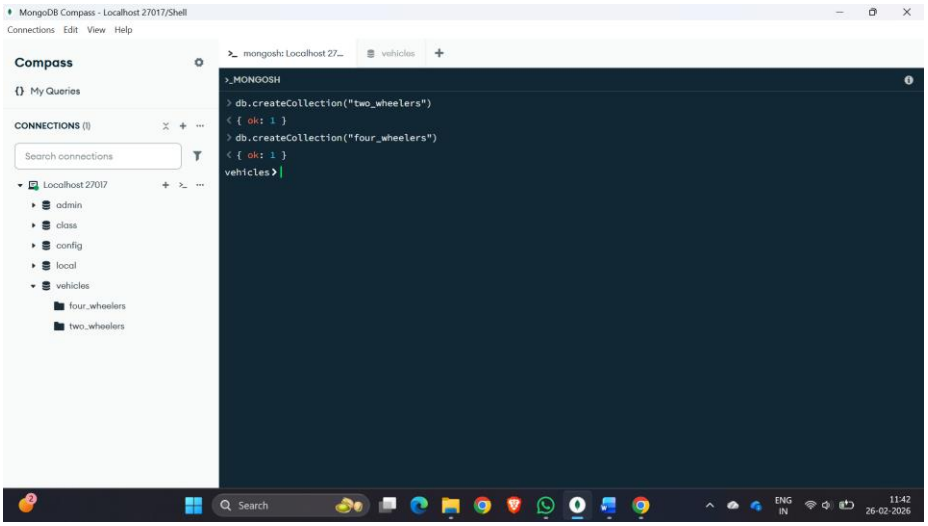
   2. Write a MongoDB query to display all the databases.

      output:



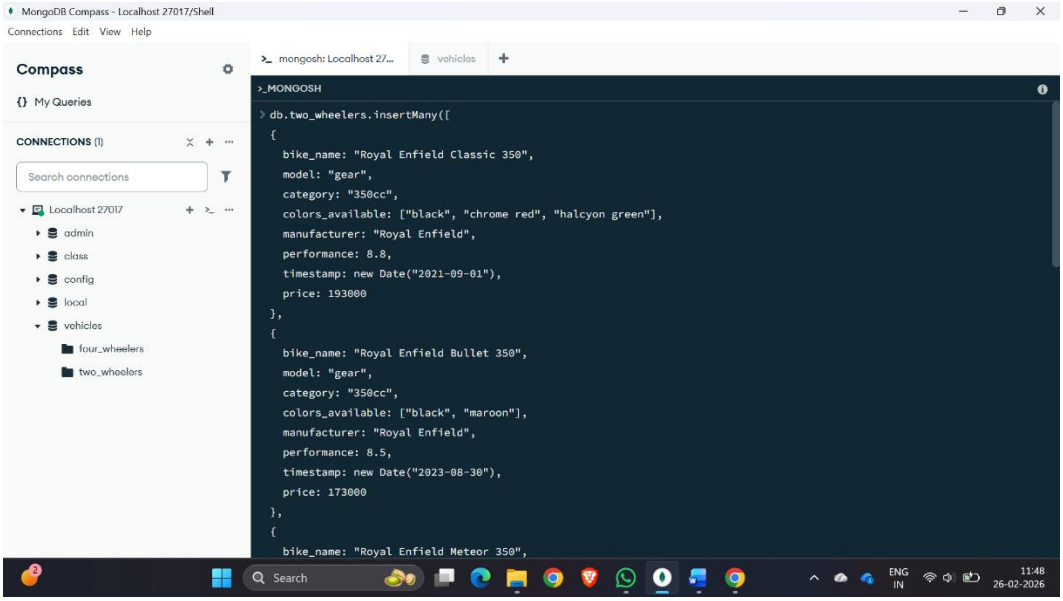
3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

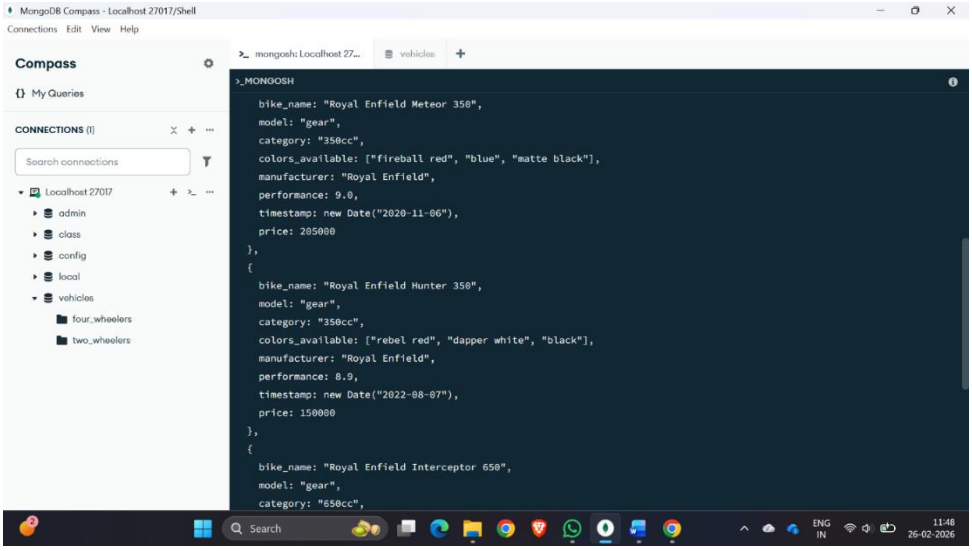
   output:

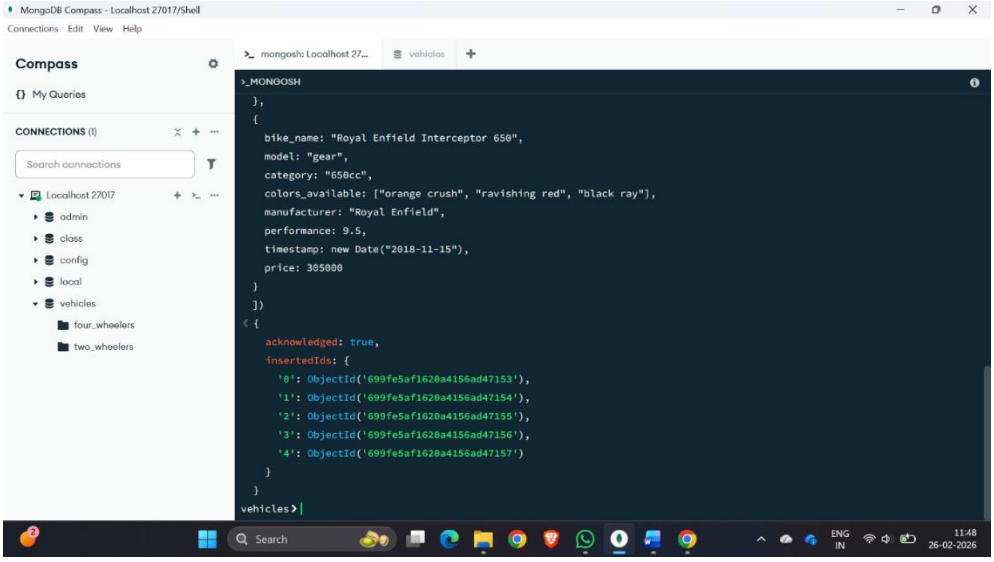


4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

   output:







5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

   output:

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017

admin

class

config

local

vehicles

four_wheelers

two_wheelers

mongosh: Localhost 27...

vehicles

>_MONGOSH

```
> db.four_wheelers.insertMany([
  {
    vehicle_name: "BMW X5",
    model: "own",
    category: "car",
    variants: ["xDrive30d", "xDrive40i", "diesel", "petrol"],
    manufacturer: "BMW",
    performance: 9.2,
    timestamp: new Date("2023-04-15"),
    price: 9500000
  },
  {
    vehicle_name: "Ferrari 488 GTB",
    model: "own",
    category: "car",
    variants: ["V8", "petrol", "automatic"],
    manufacturer: "Ferrari",
    performance: 9.8,
    timestamp: new Date("2019-02-10"),
    price: 40000000
  },
  {
    vehicle_name: "Rolls-Royce Phantom",
```

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017

admin

class

config

local

vehicles

four_wheelers

two_wheelers

mongosh: Localhost 27...

vehicles

>_MONGOSH

```
    vehicle_name: "Rolls-Royce Phantom",
    model: "own",
    category: "car",
    variants: ["V12", "petrol", "automatic"],
    manufacturer: "Rolls-Royce",
    performance: 9.7,
    timestamp: new Date("2022-01-20"),
    price: 90000000
  },
  {
    vehicle_name: "Tata Ultra 1918",
    model: "commercial",
    category: "heavy truck",
    variants: ["diesel", "manual"],
    manufacturer: "Tata Motors",
    performance: 8.5,
    timestamp: new Date("2021-07-12"),
    price: 2800000
  },
  {
    vehicle_name: "Ashok Leyland Dost",
    model: "commercial",
```

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017

admin

class

config

local

vehicles

four_wheelers

two_wheelers

mongosh: Localhost 27...

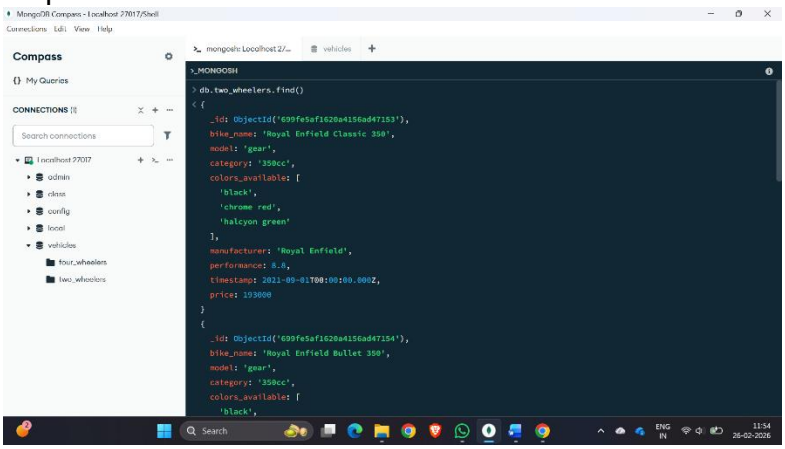
vehicles

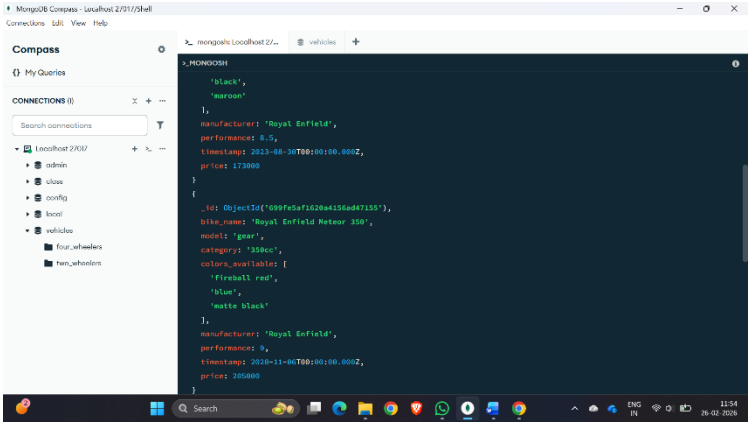
>_MONGOSH

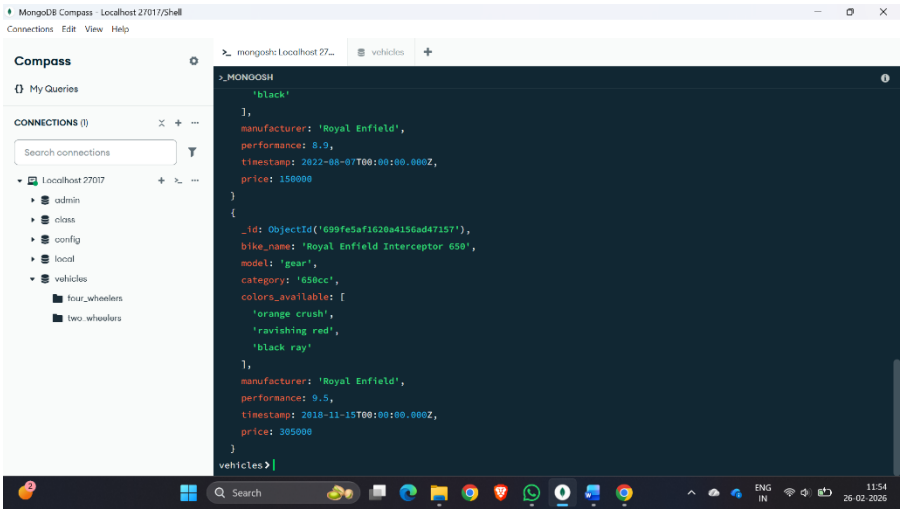
```
  },
  {
    vehicle_name: "Ashok Leyland Dost",
    model: "commercial",
    category: "mini truck",
    variants: ["diesel", "LS variant"],
    manufacturer: "Ashok Leyland",
    performance: 8.2,
    timestamp: new Date("2020-09-05"),
    price: 750000
  }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('699fe69d1620a4156ad47158'),
      '1': ObjectId('699fe69d1620a4156ad47159'),
      '2': ObjectId('699fe69d1620a4156ad4715a'),
      '3': ObjectId('699fe69d1620a4156ad4715b'),
      '4': ObjectId('699fe69d1620a4156ad4715c')
    }
  }
vehicles>
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

output:







MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017
- admin
- class
- config
- local
- vehicles
  - four_wheelers
  - two_wheelers

mongosh: Localhost 27... | vehicles

```
>_MONGOSH
> db.four_wheelers.find()
< {
    _id: ObjectId('699fe69d1620a4156ad47158'),
    vehicle_name: 'BMW X5',
    model: 'own',
    category: 'car',
    variants: [
      'xDrive30d',
      'xDrive40i',
      'diesel',
      'petrol'
    ],
    manufacturer: 'BMW',
    performance: 9.2,
    timestamp: 2023-04-15T00:00:00.000Z,
    price: 9500000
  }
  {
    _id: ObjectId('699fe69d1620a4156ad47159'),
    vehicle_name: 'Ferrari 488 GTB',
    model: 'own',
    category: 'car',
    variants: [
```

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017
- admin
- class
- config
- local
- vehicles
  - four_wheelers
  - two_wheelers

mongosh: Localhost 27... | vehicles

```
>_MONGOSH
    variants: [
      'V8',
      'petrol',
      'automatic'
    ],
    manufacturer: 'Ferrari',
    performance: 9.8,
    timestamp: 2019-02-10T00:00:00.000Z,
    price: 40000000
  }
  {
    _id: ObjectId('699fe69d1620a4156ad4715a'),
    vehicle_name: 'Rolls-Royce Phantom',
    model: 'own',
    category: 'car',
    variants: [
      'V12',
      'petrol',
      'automatic'
    ],
    manufacturer: 'Rolls-Royce',
    performance: 9.7,
    timestamp: 2022-01-20T00:00:00.000Z,
```

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

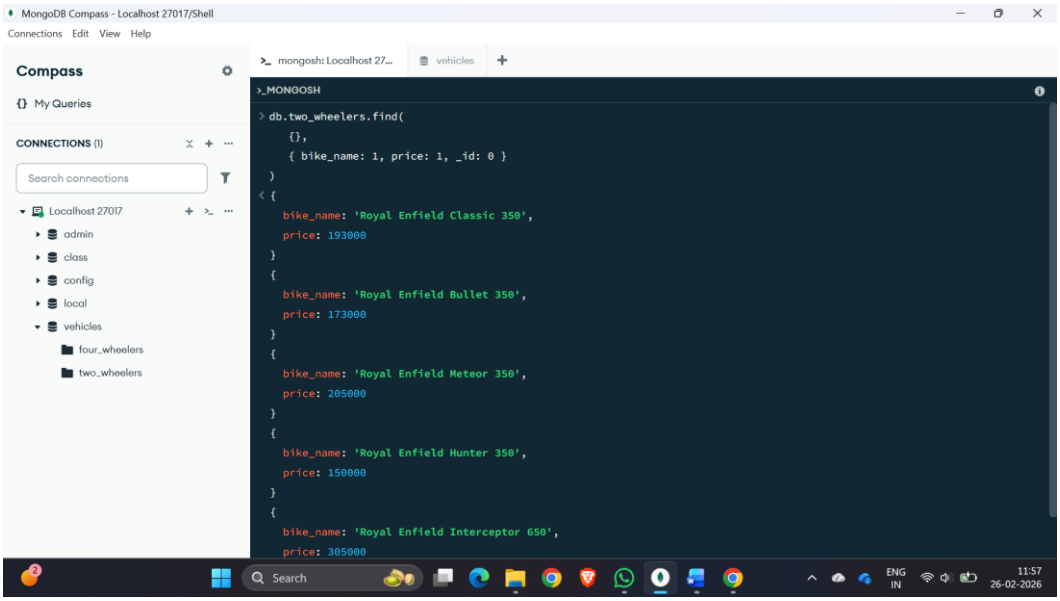
Localhost 27017
- admin
- class
- config
- local
- vehicles
  - four_wheelers
  - two_wheelers

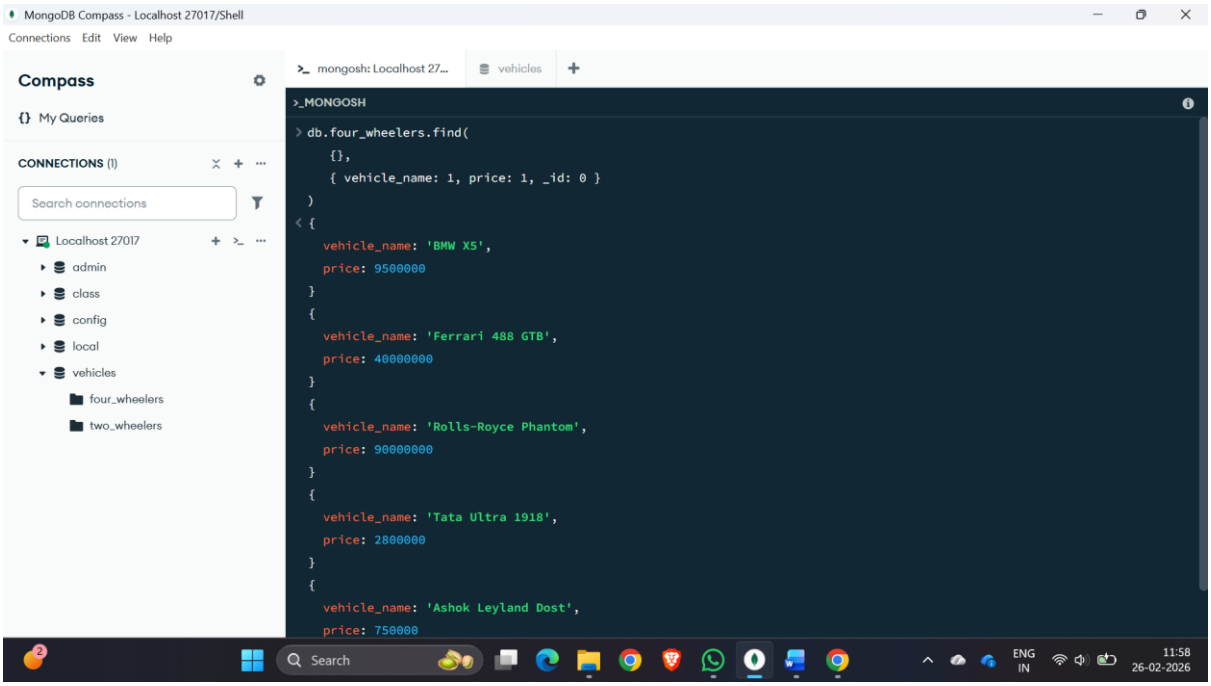
mongosh: Localhost 27... | vehicles

```
>_MONGOSH
      'diesel',
      'manual'
    ],
    manufacturer: 'Tata Motors',
    performance: 8.5,
    timestamp: 2021-07-12T00:00:00.000Z,
    price: 2800000
  }
  {
    _id: ObjectId('699fe69d1620a4156ad4715c'),
    vehicle_name: 'Ashok Leyland Dost',
    model: 'commercial',
    category: 'mini truck',
    variants: [
      'diesel',
      'LS variant'
    ],
    manufacturer: 'Ashok Leyland',
    performance: 8.2,
    timestamp: 2020-09-05T00:00:00.000Z,
    price: 750000
  }
vehicles>
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

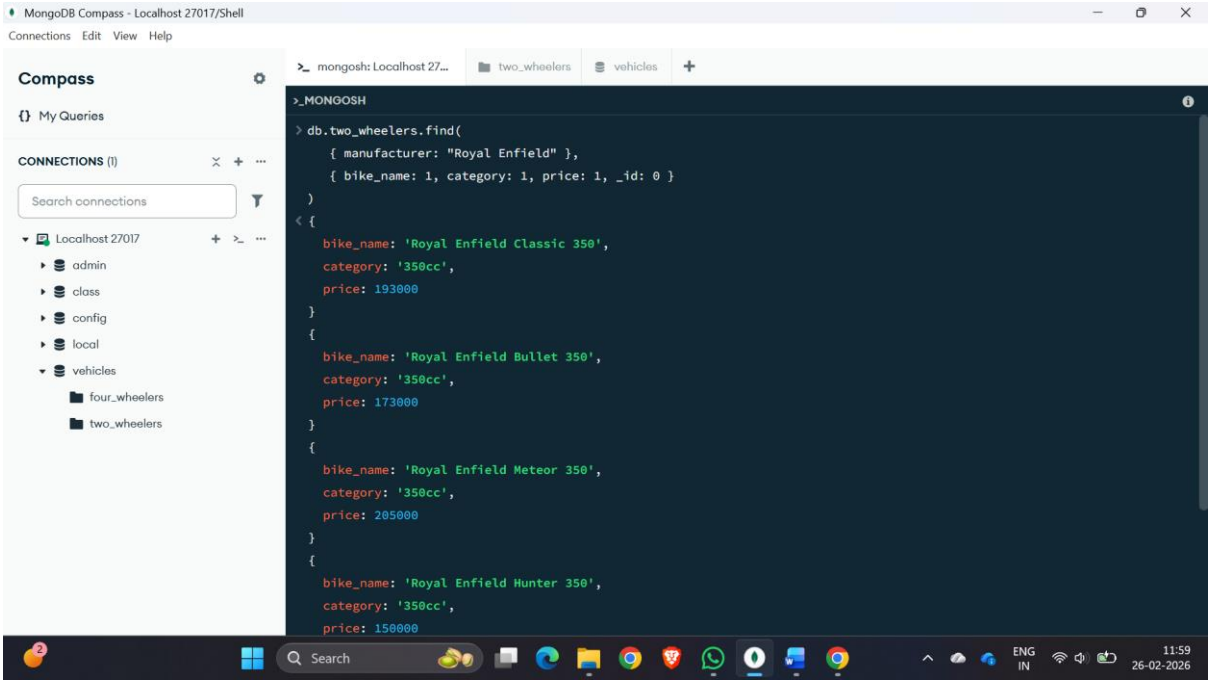
output:





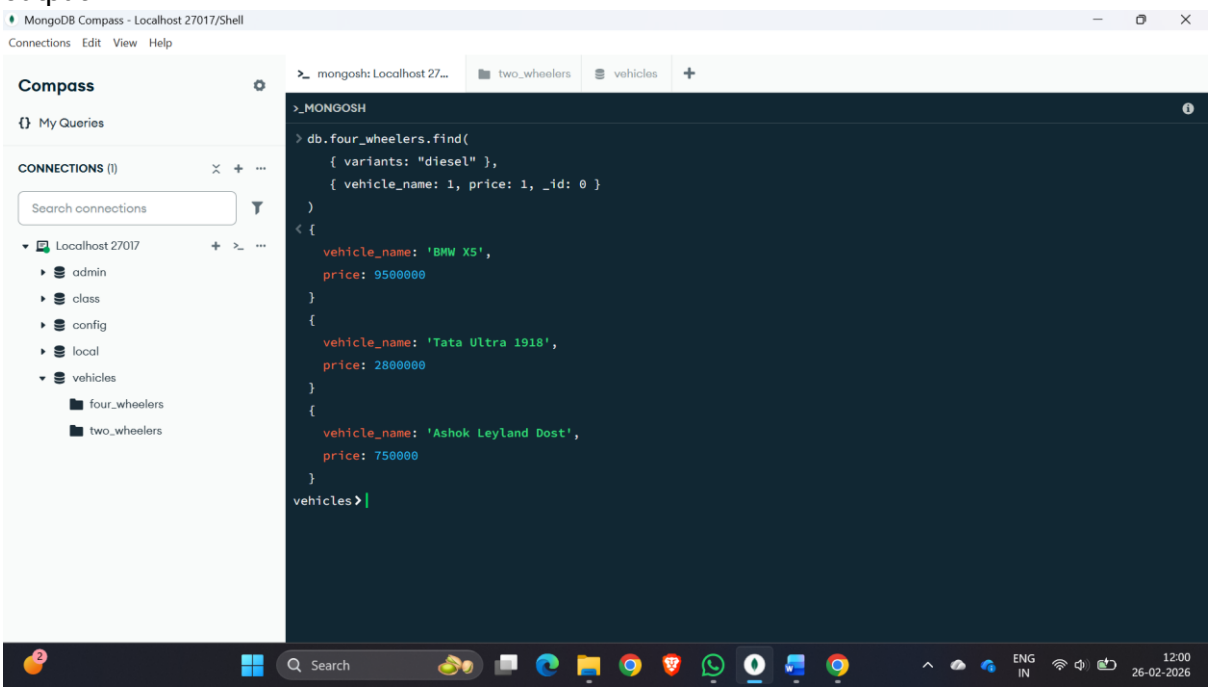
8. Write a MongoDB query to display two_wheelers from a particular company

output:



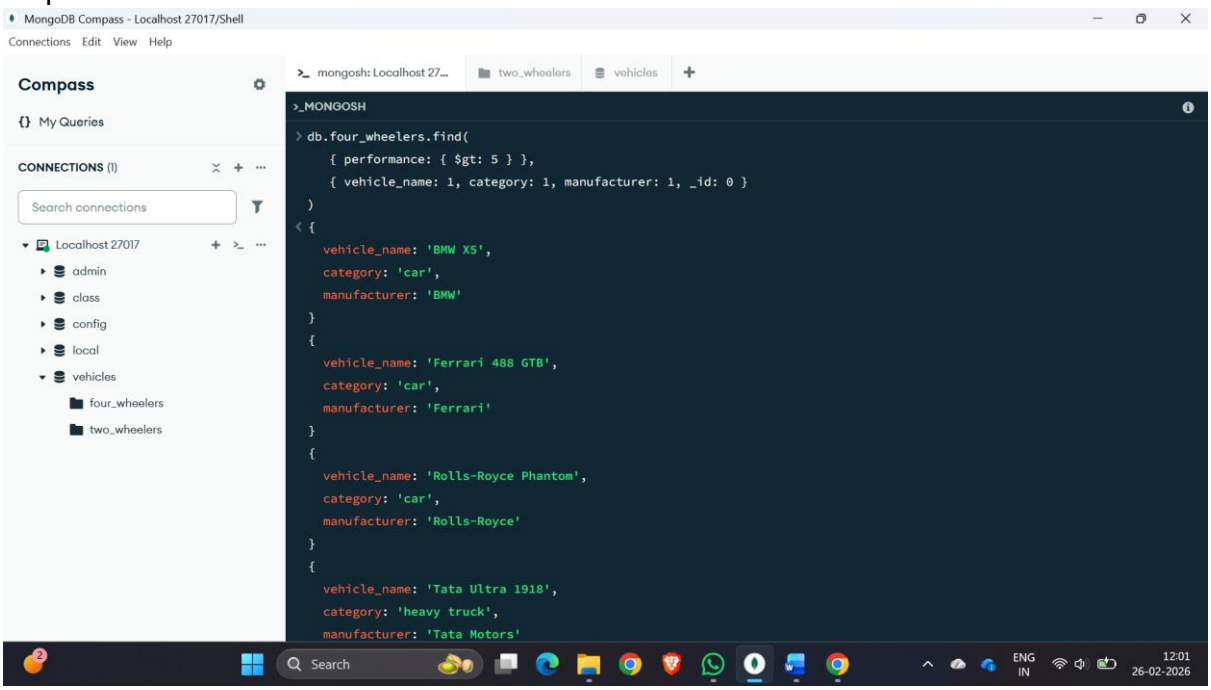
9. Write a MongoDB query to display four_wheelers available in diesel variants

output:



10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

output:



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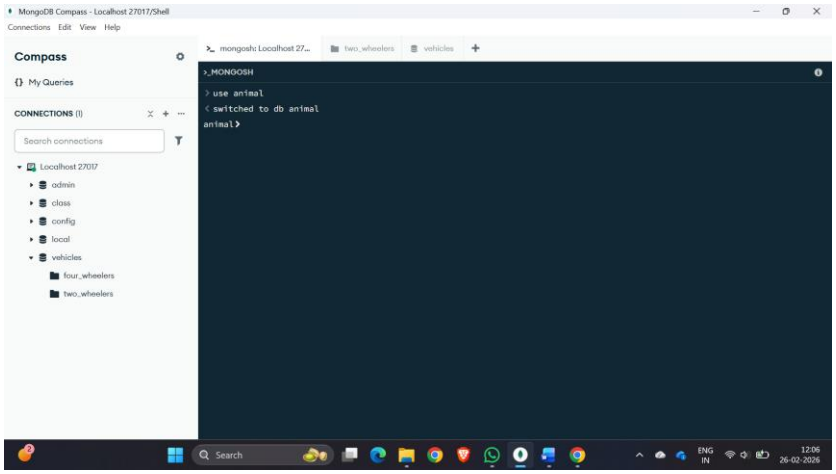
Branch/ Class: B.Tech/M.Tech **Date:26-02-2026**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: P.Prasanth kumar **Reg. no.: 23BCE9742**

2. Use MongoDB to implement the following DB operations for a Zoo

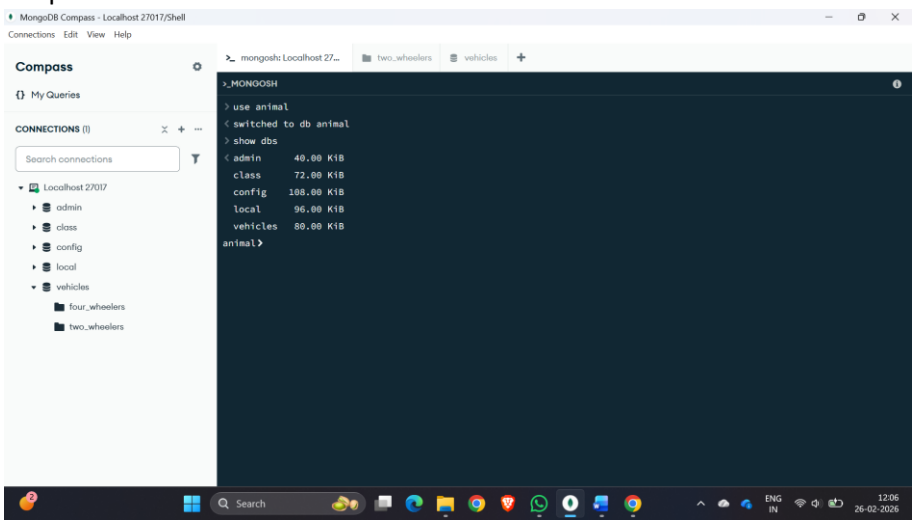
1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

   Output:-



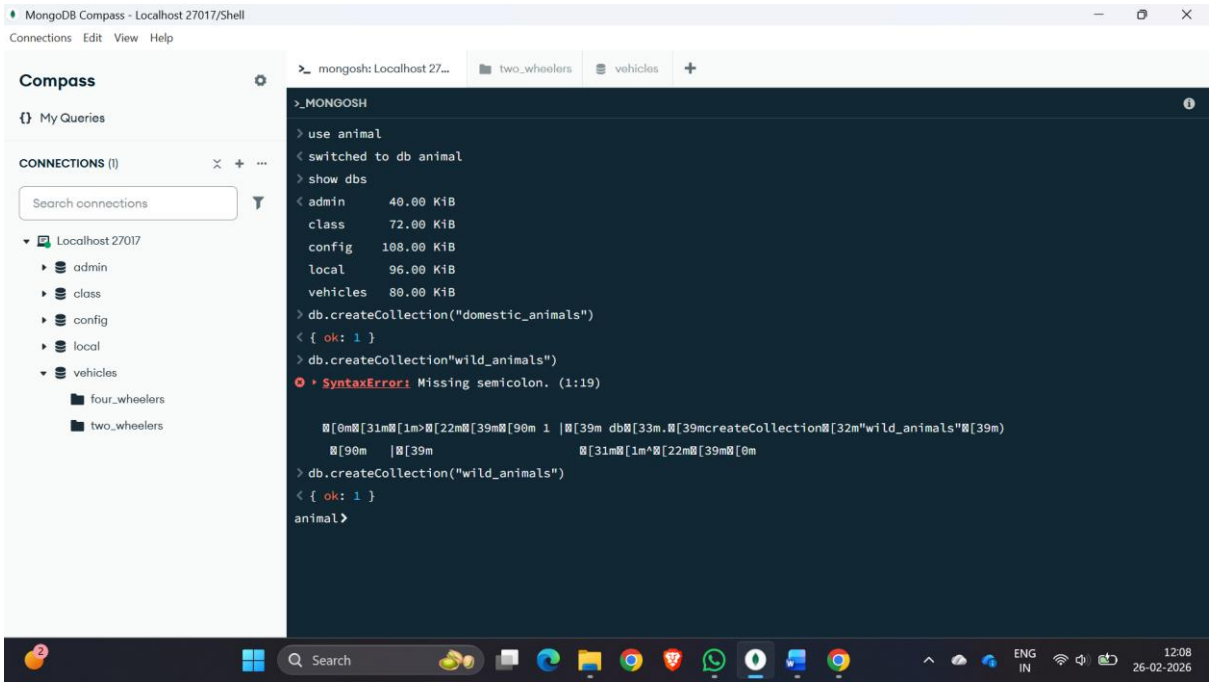
2. Write a MongoDB query to display all the databases.

   Output:-



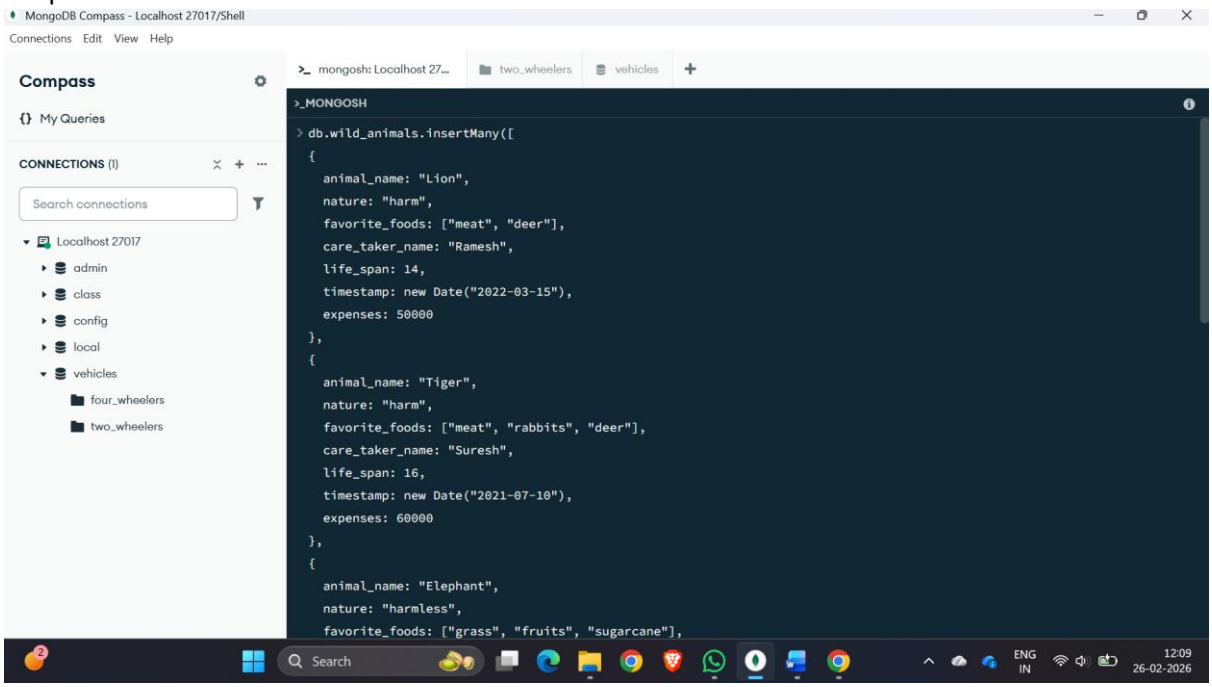
3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

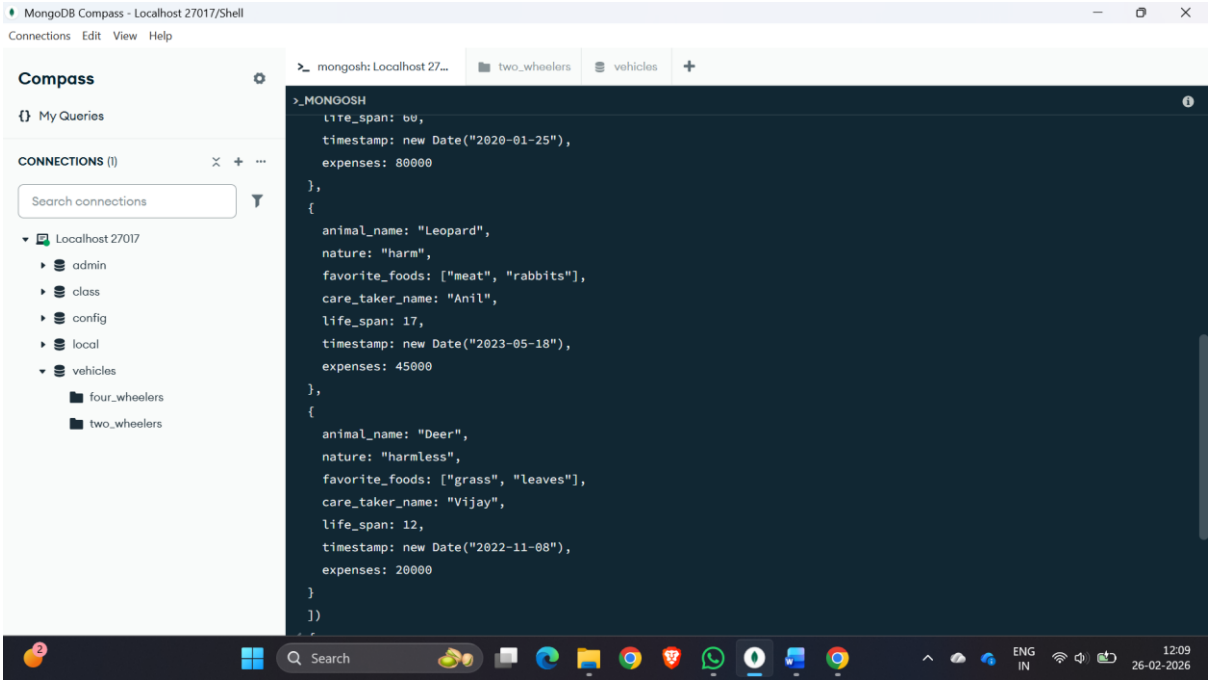
Output:-



4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

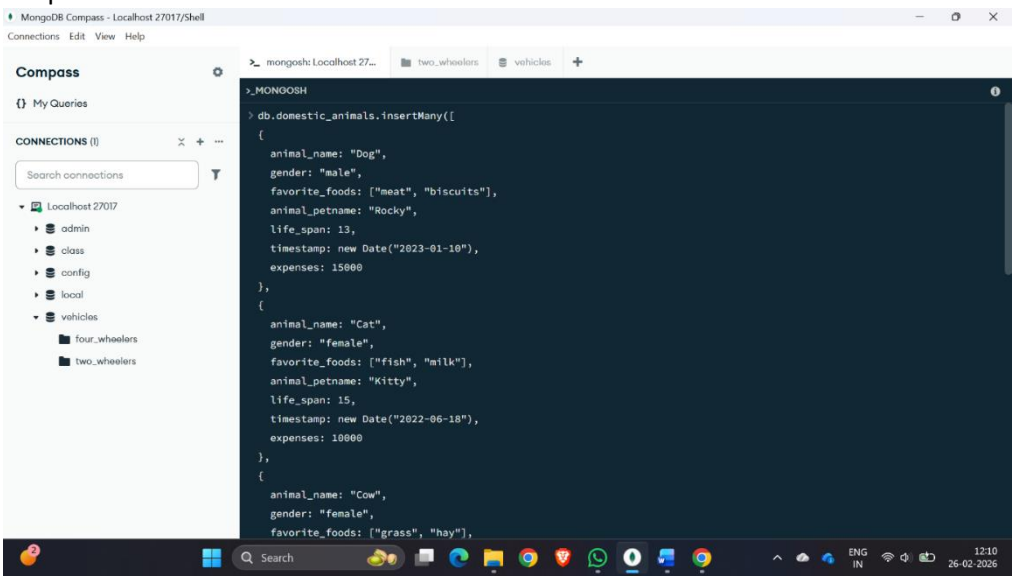
Output:-

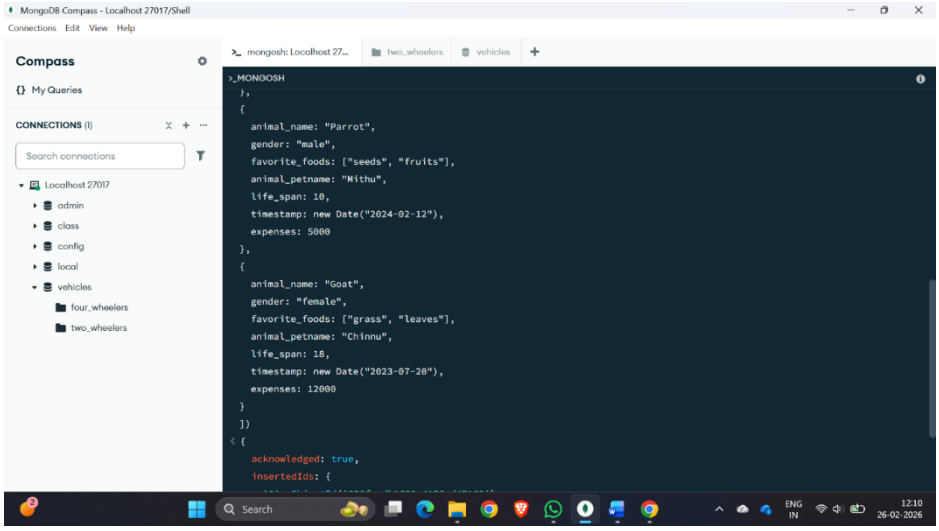




5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

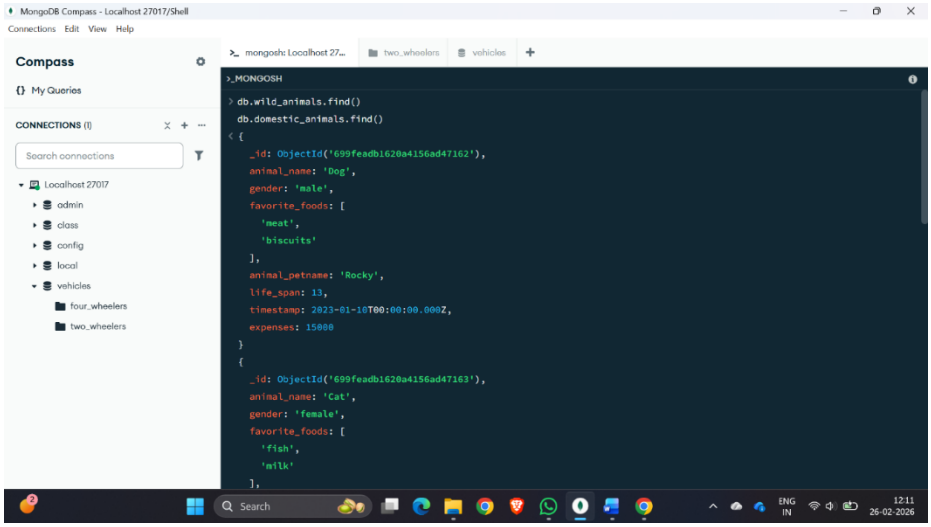
Output:-

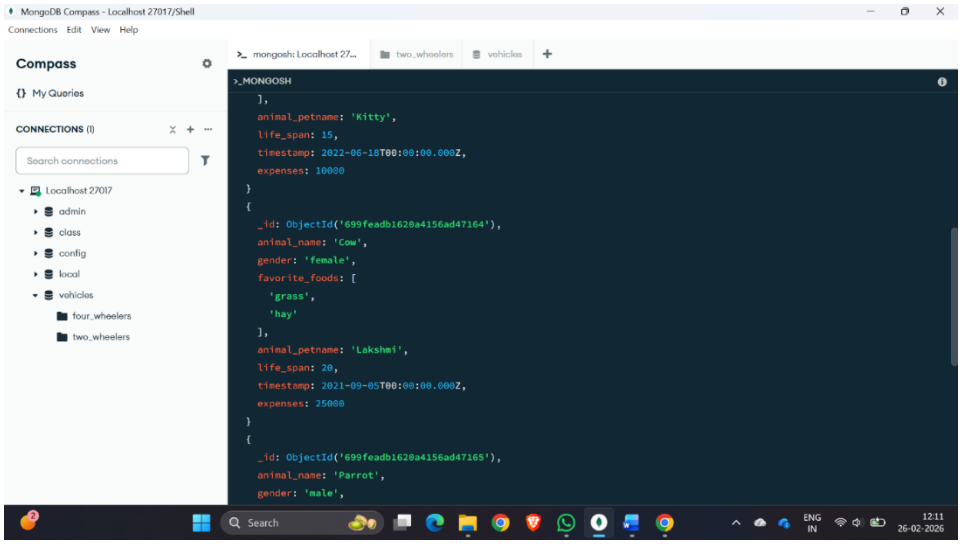


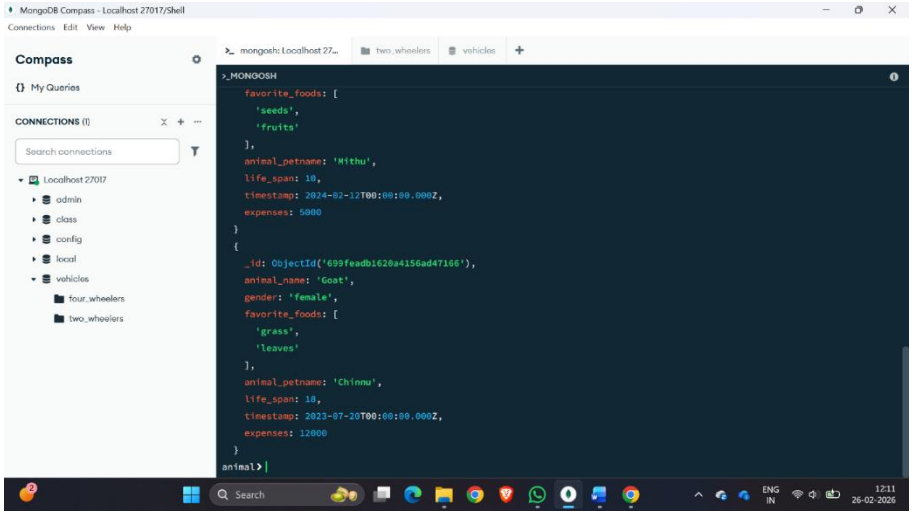


6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

Output:-

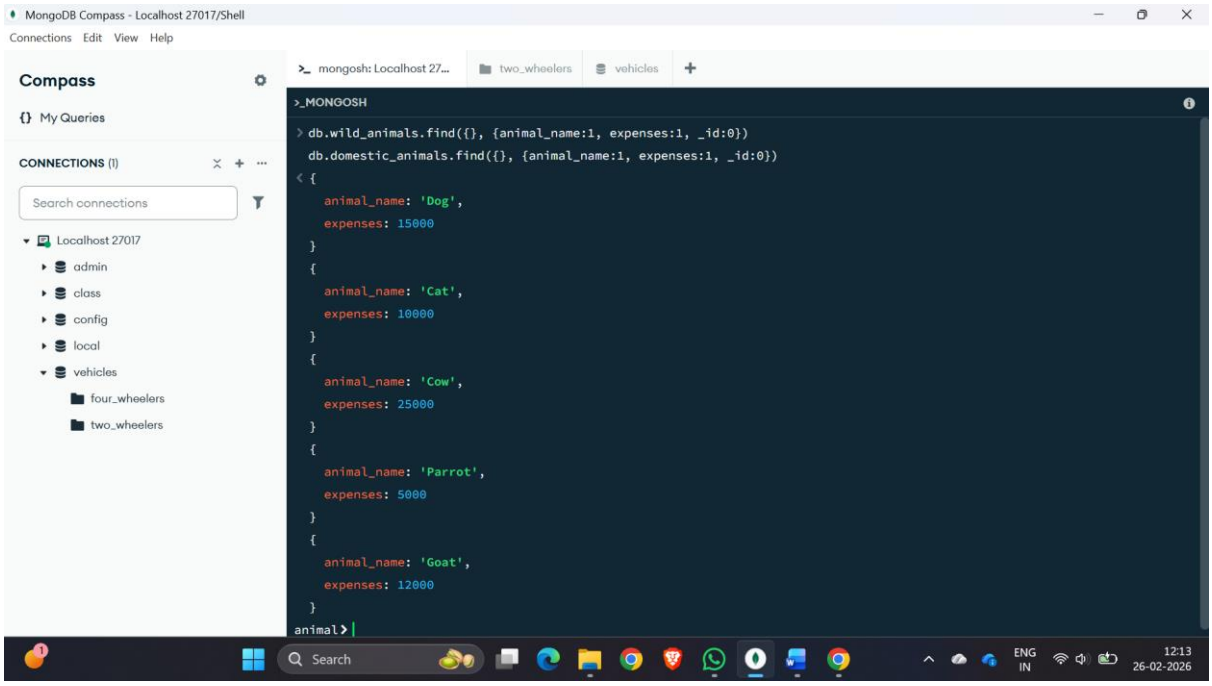






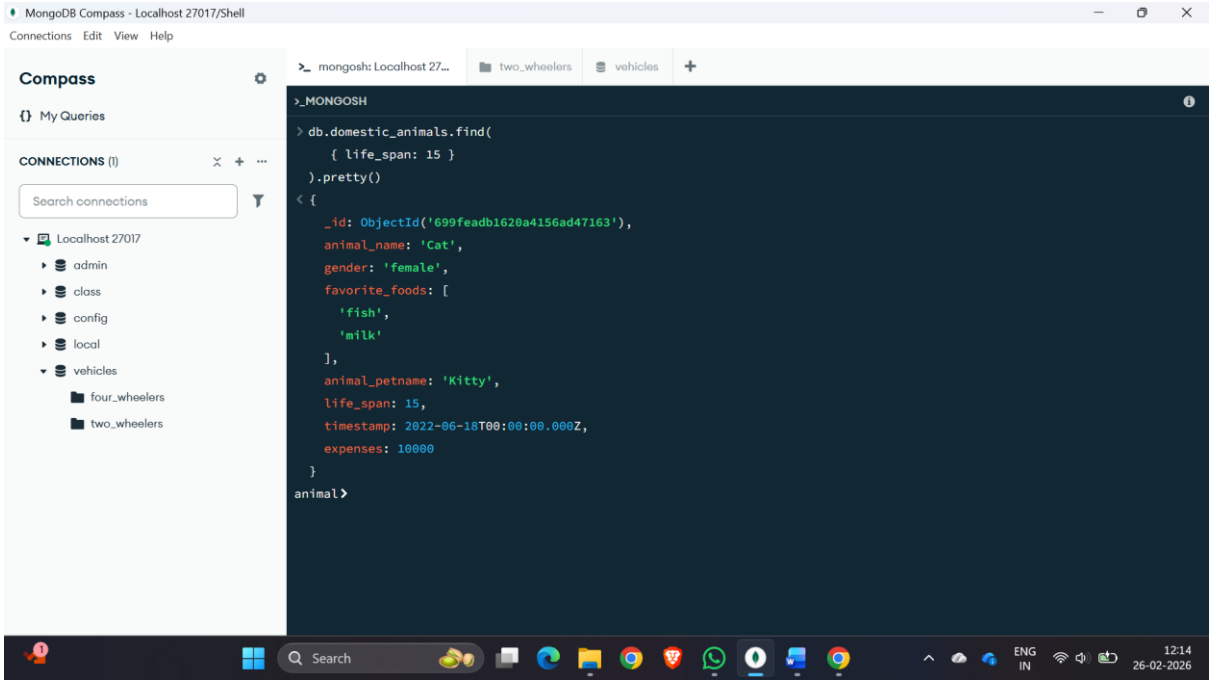
7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

Output:-



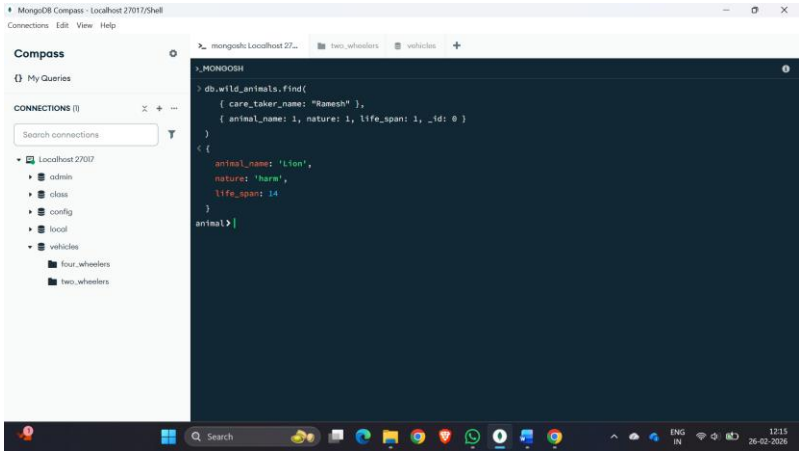
8. Write a MongoDB query to display domestic_animals whose life is a particular year

Output:-



9. Write a MongoDB query to display wild_animals available under a particular care_taker

Output:-



10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

Output:-

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017

admin

class

config

local

vehicles

four_wheelers

two_wheelers

mongosh: Localhost 27...

two_wheelers

vehicles

>_MONGOSH

```
> db.wild_animals.find({life_span:{$gt:5}}, {animal_name:1,favorite_foods:1,expenses:1,_id:0})
  db.domestic_animals.find({life_span:{$gt:5}}, {animal_name:1,favorite_foods:1,expenses:1,_id:0})
< {
    animal_name: 'Dog',
    favorite_foods: [
      'meat',
      'biscuits'
    ],
    expenses: 15000
  }
  {
    animal_name: 'Cat',
    favorite_foods: [
      'fish',
      'milk'
    ],
    expenses: 10000
  }
  {
    animal_name: 'Cow',
    favorite_foods: [
      'grass',
      'hay'
```

Search

ENG IN

12:15 26-02-2026

MongoDB Compass - Localhost 27017/Shell

Connections Edit View Help

Compass

My Queries

CONNECTIONS (1)

Search connections

Localhost 27017

admin

class

config

local

vehicles

four_wheelers

two_wheelers

mongosh: Localhost 27...

two_wheelers

vehicles

>_MONGOSH

```
    favorite_foods: [
      'grass',
      'hay'
    ],
    expenses: 25000
  }
  {
    animal_name: 'Parrot',
    favorite_foods: [
      'seeds',
      'fruits'
    ],
    expenses: 5000
  }
  {
    animal_name: 'Goat',
    favorite_foods: [
      'grass',
      'leaves'
    ],
    expenses: 12000
  }
animal>
```

Search

ENG IN

12:15 26-02-2026